1 6.S952

- (Everett, 1973)
- (Zurek, 2002)
- (Deutsch, 1985)
- (Foster & Brown, 1988)
- (Wikipedia contributors, 2026a)
- (Chaitin, 2004)
- (Einstein et al., 1935)
- (Bell, 1992)
- (Bennett et al., 1993)
- (Kwiat et al., 1996)
- (Penrose & Rindler, 1984)
- (Feynman, 2014)
- (Taylor et al., 1998)
- (Cariglia & Alves, 2015)
- (Cleve et al., 1998)
- (Ismael, 2025)
- (Albert, 1992)

2 Books

- (Bretscher, 2009)
- (Covey, 1990)
- (Watts, 1972)
- (Eliot, 1871)
- (Margolis et al., 2011)
- (Duda et al., 2001)
- (Bishop, 2006)
- (Mitchell, 1997)
- (Toffoli & Margolus, 1987)
- (Euclid, 2003)
- (Feldman, 2018)
- (Schimmer et al., 2018)

3 Papers

- (Wing, 2006)
- (Rabiner, 1989)
- (Gershenson, 2003)
- (de Campos et al., 2009c)
- (de Campos et al., 2009a)
- (de Campos et al., 2009b)

4 Software

- (Kernighan & Ritchie, 1988)
- (Sedgwick & Wayne, 2011)

5 Teaching

- (Belvel & Jordan, 2003)
- (Smith, 2004)
- (Marzano et al., 2004)

6 Mathematics

- (Larson et al., 2004b)
- (Larson et al., 2004a)
- (Tucker, 2002)
- (Devlin, 2008)
- (Davis, 2000)
- (Boole, 1847)

7 WWW

- (Bateman & Mayer, 2005)
- (Hubbard & Roby, 2008)
- (Poskanzer, 2009)
- (Goode & Chapman, 2013)
- (Chapman et al., 2025)
- (Center for Technology in Learning at SRI International, 2013)
- (Snyder et al., 2013)
- (Seehorn et al., 2011)
- (Stephenson, 2012)

8 Wikipedia

- (Wikipedia contributors, 2019s)
- (Wikipedia contributors, 2019j)
- (Wikipedia contributors, 2019b)
- (Wikipedia contributors, 2019d)
- (Wikipedia contributors, 2018a)
- (Wikipedia contributors, 2019r)
- (Wikipedia contributors, 2019c)
- (Wikipedia contributors, 2019n)
- (Wikipedia contributors, 2019a)
- (Wikipedia contributors, 2017b)
- (Wikipedia contributors, 2019g)
- (Wikipedia contributors, 2019h)
- (Wikipedia contributors, 2019p)
- (Wikipedia contributors, 2019m)
- (Wikipedia contributors, 2018b)
- (Wikipedia contributors, 2017a)
- (Wikipedia contributors, 2019f)
- (Wikipedia contributors, 2018d)
- (Wikipedia contributors, 2019t)
- (Wikipedia contributors, 2019o)
- (Wikipedia contributors, 2019u)
- (Wikipedia contributors, 2018c)
- (Wikipedia contributors, 2019k)
- (Wikipedia contributors, 2019l)
- (Wikipedia contributors, 2019i)
- (Wikipedia contributors, 2019e)
- (Wikipedia contributors, 2018e)
- (Wikipedia contributors, 2019q)
- (Wikipedia contributors, 2010a)
- (Wikipedia contributors, 2010b)
- (Wikipedia contributors, 2010c)
- (Wikipedia contributors, 2013)
- (Wikipedia contributors, 2004)
- (Wikibooks contributors, 2018)
- (Wikipedia contributors, 2023)
- (Wikipedia contributors, 2026b)

9 Debug

- (van de Donk et al., 2004)
- ("This is the title of the webpage", 2009)
- (Jech, 2011)
- (CiteDrive, 2022)

10 MIT App Inventor

10.1 2026

- (Lee et al., 2026)
- (Zamanova et al., 2026)
- (D. Kim et al., 2026)
- (Afrida et al., 2026)

10.2 2025

- (Yuan, 2025)
- (Patton et al., 2025)

10.3 2024

- (Ravi et al., 2024)
- (Higman, 2024)
- (Granquist, 2024)
- (Elhashemy, 2024)
- (D. Y. J. Kim et al., 2024)
- (D. Kim et al., 2024)

10.4 2023

- (Granquist et al., 2023)
- (Yoo et al., 2023)
- (Iniego, 2023)
- (Elhashemy et al., 2023)
- (Tian, 2023)
- (Pang et al., 2023b)
- (Pang et al., 2023a)
- (Du, Parks, et al., 2023)
- (A. Zhou et al., 2023)
- (Van Brummelen, Tian, et al., 2023)
- (Van Brummelen, Kelleher, et al., 2023)
- (Du, Taylor, et al., 2023)

10.5 2022

- (D. Y. J. Kim et al., 2022)
- (Pang et al., 2022)
- (Shone et al., 2022)
- (Adebayo, 2022)
- (Van Brummelen, 2022)
- (Shone, 2022)
- (Pang, 2022)
- (Harkavy, 2022)

10.6 2021

- (J. Zhu & Van Brummelen, 2021)
- (T.-C. Hsu, Abelson, & Van Brummelen, 2021)
- (T.-C. Hsu, Abelson, Patton, et al., 2021)
- (J. Zhu, 2021)
- (Yu, 2021)
- (Dunand, 2021)
- (Lin & Van Brummelen, 2021)
- (Van Brummelen, Tabunshchyk, & Heng, 2021)
- (Van Brummelen, Heng, & Tabunshchyk, 2021)
- (T.-C. Hsu, Abelson, Lao, et al., 2021)

10.7 2020

- (Lao, 2020)
- (Weng, 2020)
- (Van Brummelen, Weng, et al., 2020)
- (Lin et al., 2020)
- (Van Brummelen, Yeo, & Weng, 2020)
- (Bhatia, 2020)
- (Lao & Bhatia, 2020)
- (Lao et al., 2020)
- (Martin et al., 2020)
- (X. Zhou et al., 2020)

10.8 2019

- (Van Brummelen, 2019b)
- (Van Brummelen et al., 2019)
- (Patton, Seo, & Turbak, 2019)
- (Patton, 2019a)
- (Patton, 2019b)
- (D. Tang et al., 2019)
- (Lao et al., 2019)
- (Van Brummelen, 2019a)
- (Tissenbaum et al., 2019)
- (K. Zhu, 2019)
- (D. Tang, 2019)
- (Chan, 2019)
- (Van Brummelen, 2019c)
- (Clarke, 2019)
- (Patton, Tissenbaum, & Harunani, 2019)

10.9 2018

- (Tissenbaum et al., 2018)
- (Lao, 2018)
- (Patton, Druga, et al., 2018)
- (Patton, Sherman, & Tissenbaum, 2018)
- (Patton & Tang, 2018)
- (Xi & Patton, 2018)
- (Van Brummelen & Abelson, 2018)
- (Kong et al., 2018)
- (Tseng et al., 2018)
- (L. Tang, 2018)
- (Prutz, 2018)
- (Prutz & Abelson, 2018)
- (Wang, 2018)
- (Sherman & Alkhatib, 2018)

10.10 2017

- (Tissenbaum, Sheldon, Seop, et al., 2017)
- (Kong et al., 2017)
- (Ruan et al., 2017)
- (Bau et al., 2017)
- (Tissenbaum, Sheldon, Patton, et al., 2017)
- (Lao, 2017a)
- (Lao, 2017b)
- (Deng, 2017)
- (L. Tang, 2017)
- (Deng & Patton, 2017)
- (Patton & Deng, 2017)
- (Campbell, 2017)
- (Giurleo, 2017)
- (Im, 2017)

10.11 2016

- (Li, 2016)
- (Colter, 2016)
- (Xie, 2016)
- (Tsai, 2016)
- (Asmus, 2016)

10.12 2015

- (Xie et al., 2015)
- (Morelli et al., 2015)

10.13 2014

- (Turbak et al., 2014)
- (Miao, 2014)
- (Wolber et al., 2014)
- (Schiller et al., 2014)

10.14 2013

- (Shih & Boortz, 2013)
- (Zhang et al., 2013)
- (Shih et al., 2013)
- (Morelli et al., 2013)
- (Crawford Pokress & Dominguez Veiga, 2013)
- (Shih & Li, 2013)
- (Y.-C. Hsu & Ching, 2013)

10.15 2012

- (Bhagi, 2012)
- (Turbak et al., 2012)
- (Margulieux et al., 2012)

10.16 2011

- (Saigal & Saigal, 2011)

10.17 2010

- (Magnuson, 2010)
- (Saigal, 2010)

References

Adebayo, J. (2022). *Towards effective tools for debugging machine learning models* [Doctoral dissertation, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/AdebayoJulius_PhD_thesis.pdf

Afrida, A., Patton, E., & Abelson, H. (2026). AI-driven scaffolding for novice app developers using RAG-finetuned chatbot. In H. Degen & S. Ntoa (Eds.), *Artificial intelligence in HCI* (pp. 375–387, Vol. 16746). Springer, Cham. https://doi.org/10.1007/978-3-032-31048-4_23

Albert, D. Z. (1992). *Quantum mechanics and experience*. Harvard University Press. Retrieved September 9, 2026, from https://isbn.nu/9780674741126

Asmus, K. (2016). *A system of applications for the integration of BLE beacons in museums* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B3jsksMcCW5bVGZ6WGdJWjUxM1E

Bateman, W., & Mayer, S. (2005). The online LaTeX equation editor. Retrieved October 10, 2026, from https://editor.codecogs.com/

Bau, D., Gray, J., Kelleher, C., Sheldon, J., & Turbak, F. (2017). Learnable programming: Blocks and beyond. *Communications of the ACM*, *60*(6), 72–80. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B8AUNoeCnCCAbGVBcUhSNnZDZVk

Bell, J. S. (1992). Six possible worlds of quantum mechanics [Originally delivered as an essay at Nobel Symposium 65 (1986) and first published in *Possible Worlds in Humanities, Arts and Sciences* (1989); this is the 1992 journal republication]. *Foundations of Physics*, *22*(10), 1201–1215. Retrieved September 30, 2026, from https://personal.lse.ac.uk/ROBERT49/teaching/partiii/pdf/BellSixPossibleWorldsQmMechsFdnsPhysics1992.pdf

Belvel, P. S., & Jordan, M. M. (2003). *Rethinking classroom management: Strategies for prevention, intervention, and problem solving*. Corwin Press.

Bennett, C. H., Brassard, G., Crépeau, C., Jozsa, R., Peres, A., & Wootters, W. K. (1993). Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels

[Original publisher's copy is paywalled at https://journals.aps.org/prl/pdf/10.1103/PhysRevLett.70.1895]. *Physical Review Letters*, *70*(13), 1895–1899. https://doi.org/10.1103/PhysRevLett.70.1895

Bhagi, A. (2012). *Android game development with App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/explore/sites/all/files/Resources/Thesis_FINAL_AnshulBhagi.pdf

Bhatia, N. (2020). *Using transfer learning, spectrogram audio classification, and MIT App Inventor to facilitate machine learning understanding* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Nikhil_Bhatia_MEng_Thesis.pdf

Bishop, C. M. (2006). *Pattern recognition and machine learning*. Springer Science+Business Media.

Boole, G. (1847). *The mathematical analysis of logic: Being an essay towards a calculus of deductive reasoning*. Macmillan, Barclay, & Macmillan.

Bretscher, O. (2009). *Linear algebra with applications* (4th). Pearson Prentice Hall. https://isbn.nu/9780136009269

Campbell, G. (2017). *Extending cloud data store component for MIT App Inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B8AUNoeCnCCAbl9iVnJQYnoyaGc

Cariglia, M., & Alves, F. K. (2015). The eisenhart lift: A didactical introduction of modern geometrical concepts from hamiltonian dynamics. *European Journal of Physics*, *36*(2), 025018. https://doi.org/10.1088/0143-0807/36/2/025018

Center for Technology in Learning at SRI International. (2013). Draft assessment design patterns, ECS computational thinking practices. Retrieved August 10, 2013, from http://pact.sri.com/?page_id=832

Chaitin, G. (2004). How real are real numbers? *CoRR*, *math.HO/0411418*. https://arxiv.org/abs/math/0411418v3

Chan, K. K. (2019). *Building an online community of creators through MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=19XiYfawHkFjxDVKgbK9k1hP5VDOjIC5F

Chapman, G., Goode, J., & Jayathirtha, G. (2025). Exploring computer science [© University of Oregon, 2025; labeled a 'teacher copy' intended solely for individual classroom use; do not copy or distribute without written permission from https://exploringcs.org/]. Retrieved September 15, 2026, from https://www.exploringcs.org/wp-content/uploads/2025/09/ECS-v10.pdf

CiteDrive, I. (2022). *Citedrive brings reference management to overleaf*. Retrieved August 12, 2023, from https://www.citedrive.com/overleaf

Clarke, N. I. (2019). *Through the screen and into the world: Augmented reality components with MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://dspace.mit.edu/handle/1721.1/123016

Cleve, R., Ekert, A., Macchiavello, C., & Mosca, M. (1998). Quantum algorithms revisited. *Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences*, *454*(1969), 339–354. https://doi.org/10.1098/rspa.1998.0164

Colter, A. (2016). *Evaluating and improving the usability of MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B3jsksMcCW5bbGw4ZHZvMGQ0bVk

Covey, S. R. (1990). *The 7 habits of highly effective people*. Free Press.

Crawford Pokress, S., & Dominguez Veiga, J. J. (2013). *MIT App Inventor: Enabling personal mobile computing* [Presented at Programming for Mobile and Touch (PRoMoTo), Indianapolis, IN]. Retrieved September 15, 2026, from http://arxiv.org/abs/1310.2830

Davis, M. (2000). *Engines of logic: Mathematics and the origin of the computer*. W. W. Norton & Company.

de Campos, T. E., Babu, B. R., & Varma, M. (2009a). The chars74k dataset. Retrieved June 27, 2010, from https://www.ee.surrey.ac.uk/CVSSP/demos/chars74k/

de Campos, T. E., Babu, B. R., & Varma, M. (2009b). Chars74k hand-drawn characters. Retrieved June 27, 2010, from https://www.ee.surrey.ac.uk/CVSSP/demos/chars74k/EnglishHnd.tgz

de Campos, T. E., Babu, B. R., & Varma, M. (2009c). Character recognition in natural images. *Proceedings of the International Conference on Computer Vision Theory and Applications, Lisbon, Portugal*. Retrieved October 11, 2026, from https://www.scitepress.org/papers/2009/17701/17701.pdf

Deng, X. (2017). *Group collaboration with App Inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B8AUNoeCnCCANHI4X1RkLXh5RTg

Deng, X., & Patton, E. W. (2017). Enabling multi-user computational thinking with collaborative blocks programming in MIT App Inventor. *Proceedings of the 1st International Conference on Computational Thinking in Education*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/CTE_2017_paper_84.pdf

Deutsch, D. (1985). Quantum theory as a universal physical theory. *International Journal of Theoretical Physics*, *24*(1), 1–41. https://doi.org/10.1007/BF00670071

Devlin, K. (2008). *The unfinished game: Pascal, Fermat, and the seventeenth-century letter that made the world modern*. Basic Books. https://isbn.nu/9780465009107

Du, X., Parks, R., Tezel, S., Freilich, J., Pang, H. N., Abelson, H., & Breazeal, C. (2023). Designing a computational action program to tackle global challenges. *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 2*, 1320. https://doi.org/10.1145/3545947.3576267

Du, X., Taylor, M., Blumofe, N., Berman, E., Lewallen, N., Geter, S., Park, H. W., Klopfer, E., Abelson, H., & Breazeal, C. (2023). Widening the global access of artificial intelligence (AI) literacy curriculum through the participation of day of AI [Listed as "To Appear"]. *SITE 23 Proceedings*.

Duda, R. O., Hart, P. E., & Stork, D. G. (2001). *Pattern classification*. Springer Science+Business Media.

Dunand, M. (2021). *Developing visual accessibility options to empower grade school students in designing inclusive mobile applications* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/DunandMEngThesis.pdf

Einstein, A., Podolsky, B., & Rosen, N. (1935). Can quantum-mechanical description of physical reality be considered complete? [Original publisher's copy is paywalled at https://journals.aps.org/pr/abstract/10.1103/PhysRev.47.777]. *Physical Review*, *47*(10), 777–780. https://doi.org/10.1103/PhysRev.47.777

Elhashemy, H. (2024). *From curious kids to data wizards: An empowerment approach to data science education* [Master's thesis, School of Computation, Information and Technology, Technical University of Munich]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/ElhashemyMSthesis.pdf

Elhashemy, H., Parks, R., Kim, D. Y. J., Patton, E., & Abelson, H. (2023). Empowering learners with a low-barrier mobile data science toolkit. *Constructionism/FabLearn 2023 Proceedings*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Elhashemy_FabL-Constr2023.pdf

Eliot, G. (1871). *Middlemarch*. Penguin Books. https://isbn.nu/9780141439549

Euclid. (2003). *Euclid's elements: All thirteen books complete in one volume* (D. Densmore, Ed.; T. L. Heath, Trans.; 2nd ed.). Green Lion Press. https://isbn.nu/9781888009194

Everett, H., III. (1973). The theory of the universal wave function [Chapter 1 (Introduction)]. In B. S. DeWitt & N. Graham (Eds.), *The many-worlds interpretation of quantum mechanics* (pp. 3–140). Princeton University Press. https://doi.org/10.1515/9781400868056

Feldman, J. (2018). *Grading for equity: What it is, why it matters, and how it can transform schools and classrooms*. SAGE Publication. https://isbn.nu/9781506391571

Feynman, R. P. (2014). *QED: The strange theory of light and matter* [Introduction by A. Zee; Princeton Science Library edition; originally published 1988]. Princeton University Press. Retrieved September 23, 2026, from https://press.princeton.edu/books/paperback/9780691164090/qed

Foster, S., & Brown, H. (1988). On a recent attempt to define the interpretation basis in the many worlds interpretation of quantum mechanics. *International Journal of Theoretical Physics*, *27*(12), 1507–1531. https://doi.org/10.1007/BF00669289

Gershenson, C. (2003). Artificial neural networks for beginners. *CoRR*, *cs.NE/0308031*. https://arxiv.org/abs/cs.NE/0308031

Giurleo, E. (2017). *Keyboard interactions in block-based coding environments* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B8AUNoeCnCCASzUwMWtwWUhmN1E

Goode, J., & Chapman, G. (2013). Exploring computer science. Retrieved August 5, 2013, from https://www.exploringcs.org/wp-content/uploads/2010/08/ExploringComputerScience-v5.0.pdf

Granquist, A. M. (2024). *AI-augmented interface for incremental app development in MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology] [Also deposited at https://dspace.mit.edu/e e244-4619-8199-52bf4f150c35]. Retrieved September 15, 2026, from https://dspace.mit.edu/handle/1721.1/157025

Granquist, A. M., Kim, D. Y. J., & Patton, E. (2023). AI augmented feature to edit and design mobile applications. *MobileHCI 2023 Proceedings*. Retrieved September 15, 2026, from https://daki7711.github.io/files/granquist2023aiaugmented.pdf

Harkavy, E. (2022). *Accessible AI that's out of this world: Globalizing AI literacy through problem-based learning and deep learning models in a low code environment* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://dspace.mit.edu/entities/publication/323c8fd2-2219-44a3-a760-87167d2ac700

Higman, M. (2024). *Designing and writing a JSON AST interpreter for more accessible and extensible IoT communication* (Research Paper). Research Science Institute. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Higman_Miles_Marissa_Final.pdf

Hsu, T.-C., Abelson, H., Lao, N., Tseng, Y.-H., & Lin, Y.-T. (2021). Behavioral-pattern exploration and development of an instructional tool for young children to learn AI. *Computers and Education: Artificial Intelligence*, *2*, 100012. https://doi.org/10.1016/j.caeai.2021.100012

Hsu, T.-C., Abelson, H., Patton, E., Chen, S.-C., & Chang, H.-N. (2021). Self-efficacy and behavior patterns of learners using a real-time collaboration system developed for group programming. *International Journal of Computer-Supported Collaborative Learning*, *16*(4), 559–582. https://doi.org/10.1007/s11412-021-09357-3

Hsu, T.-C., Abelson, H., & Van Brummelen, J. (2021). The effects of applying experiential learning into the conversational AI learning platform on secondary school students [Special Issue "AI e-Learning and Online Curriculum"; accepted manuscript / preview version]. *International Review of Research in Open and Distributed Learning (IRRODL)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/JVB_IRRODL_manuscript.docx

Hsu, Y.-C., & Ching, Y.-H. (2013). Mobile app design for teaching and learning: Educators' experiences in an online graduate course. *The International Review of Research in Open and Distributed Learning*, *14*(4), 117–139. https://doi.org/10.19173/irrodl.v14i4.1542

Hubbard, M., & Roby, T. (2008). History. Retrieved June 27, 2010, from https://binomial.csueastbay.edu/History.html

Im, J. (2017). *User-defined blocks for MIT App Inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B8AUNoeCnCCAa3lIVkphbkVid3c

Iniego, A. N. V. (2023). *Leveraging generative AI in college application essay preparation* (Research Paper). Research Science Institute. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Iniego_AngelNicole_Shuvom_Final.pdf

Ismael, J. (2025). Quantum Mechanics. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Spring 2025). Metaphysics Research Lab, Stanford University. Retrieved September 9, 2026, from https://plato.stanford.edu/cgi-bin/encyclopedia/archinfo.cgi?entry=qm

Jech, T. (2011). Set theory. In E. N. Zalta (Ed.), *The stanford encyclopedia of philosophy* (Winter 2011). Retrieved April 23, 2014, from http://plato.stanford.edu/archives/win2011/entries/set-theory/

Kernighan, B. W., & Ritchie, D. M. (1988). *The C programming language* (Second). Prentice-Hall, Inc.

Kim, D., Bagiati, A., Mishra, T., Alghowinem, S., Taylor, M., Burghleh, T., Freilich, J., Tezel, S., Parks, R., & Breazeal, C. (2026–October 14). Futuremakers: An innovative approach to online AI education [Accepted paper]. *Proceedings of IEEE Frontiers in Education 2026 (FIE 2026)*.

Kim, D., Zhou, A., Sudo, Y., & Takano, K. (2024). Advancing mobile app development and generative AI education through MIT App Inventor. *CTE-STEM 2024 Proceedings*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/kim2024advancing.pdf

Kim, D. Y. J., Granquist, A. M., Patton, E., Friedman, M., & Abelson, H. (2022). Speak your mind: Introducing APTLY, the software platform that turns ideas into working apps. *iCERi 2022 Proceedings*. Retrieved September 15, 2026, from https://daki7711.github.io/files/ICERIpaper2022.pdf

Kim, D. Y. J., Ravi, P., Williams, R., & Yoo, D. (2024). App planner: Utilizing generative AI in K-12 mobile app development education. *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference*, 770–775. https://doi.org/10.1145/3628516.3659392

Kong, S. C., Abelson, H., Sheldon, J., Lao, A., Tissenbaum, M., & Lai, M. (2017). Curriculum activities to foster primary school students' computational practices in block-based programming environments. *International Conference on Computational Thinking Education 2017 (CTE 2017)*, 84.

Kong, S. C., Andone, D., Biswas, G., Crick, T., Hoppe, H. U., Hsu, T. C., Huang, R. H., Li, K. Y., Looi, C. K., Milrad, M., Sheldon, J., Shih, J. L., Sin, K. F., Tissenbaum, M., & Vahrenhold, J. (2018). On tools that support the development of computational thinking skills: Some thoughts and future vision [Editors' foreword to the CTE2018 proceedings volume]. In S. C. Kong, D. Andone, G. Biswas, T. Crick, H. U. Hoppe, T. C. Hsu, R. H.

Huang, K. Y. Li, C. K. Looi, M. Milrad, J. Sheldon, J. L. Shih, K. F. Sin, M. Tissenbaum, & J. Vahrenhold (Eds.), *Proceedings of the international conference on computational thinking education 2018*. The Education University of Hong Kong. Retrieved September 15, 2026, from https://drive.google.com/open?id=18MMtXBfghN6_nlAl47pCCrxQHwCsLXfG

Kwiat, P., Weinfurter, H., & Zeilinger, A. (1996). Quantum seeing in the dark [Original publisher's copy is paywalled at https://www.scientificamerican.com/article/quantum-seeing-in-the-dark/]. *Scientific American*, *275*(5), 72–78. Retrieved September 30, 2026, from https://personal.utdallas.edu/~son051000/chem3322/sci_amer_interactionfree.pdf

Lao, N. (2017a). *CloudDB: Components for exploring shared data with MIT App Inventor* [Describes ongoing development of the CloudDB component]. Retrieved September 15, 2026, from https://drive.google.com/open?id=1pJa_0gbFCWazcFVBfJXHGbTyjdjuJmXA

Lao, N. (2017b). *Developing cloud and shared data capabilities to support primary school students in creating mobile applications that affect their communities* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B8AUNoeCnCCAVDZxV2I3NS1qbHc

Lao, N. (2018). CloudDB in MIT App Inventor [Demo abstract]. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*.

Lao, N. (2020). *Reorienting machine learning education towards tinkerers and ML-engaged citizens* [Doctoral dissertation, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/NatalieLao_PhD_Dissertation.pdf

Lao, N., & Bhatia, N. (2020). Using transfer learning, spectrogram audio classification, and MIT App Inventor to facilitate machine learning understanding. *International Conference on Computational Thinking Education 2020 (CTE2020)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/CTE2020_paper_38.pdf

Lao, N., Lee, I., & Abelson, H. (2019). A deep learning practicum: Concepts and practices for teaching actionable machine learning at the tertiary education level. *ICERI 2019 Conference*. Retrieved September 15, 2026, from https://drive.google.com/open?id=13N_uS1wFbxhMzER0D8GPZf-ksB8eKomp

Lao, N., Lee, I., & Abelson, H. (2020). Experiences from teaching actionable machine learning at the university level through a small practicum approach. *International Conference on Computational Thinking Education 2020 (CTE2020)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/CTE2020_paper_9.pdf

Larson, R., Boswell, L., & Stiff, L. (2004a). *Geometry*. McDougal Littell.

Larson, R., Boswell, L., & Stiff, L. (2004b). *Geometry: Concepts and skills*. Houghton Mifflin School.

Lee, V. R., Madaio, M., Garside, B., Welch, A., Blair, K. P., Adisa, I. O., Harris, A., Holst, K., Rafael, L. B., Morad, R. L., Travis, B., Moller, B., Shields, A., Brown, Z., Hinx, L., Diaz, M., Patton, E., Tezel, S., Parks, R., ... Roschelle, J. (2026–June 19). Google, AI literacy, and the learning sciences: Multiple modes of research, industry, and practice partnerships. *International Society of the Learning Sciences (ISLS 2026)*. Retrieved September 15, 2026, from https://arxiv.org/abs/2604.07601

Li, W. (2016). *Helping the helpers: A toolkit for mobile humanitarian assistance apps* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B3jsksMcCW5bMFdKLWh2T1I3LWc

Lin, P., & Van Brummelen, J. (2021). Engaging teachers to co-design integrated AI curriculum for K-12 classrooms. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 239:1–239:12. Retrieved September 15, 2026, from https://arxiv.org/abs/2009.11100

Lin, P., Van Brummelen, J., Lukin, G., Williams, R., & Breazeal, C. (2020). Zhorai: Designing a conversational agent for children to explore machine learning concepts. *Tenth Symposium on Educational Advances in Artificial Intelligence (EAAI)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/JessVBPublications/zhorai_conversational_agent_ML.pdf

Magnuson, B. (2010). *Building blocks for mobile games: A multiplayer framework for App Inventor for Android* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from http://dspace.mit.edu/handle/1721.1/61253

Margolis, J., Estrella, R., Goode, J., Holme, J. J., & Nao, K. (2011). *Stuck in the shallow end*. MIT Press. https://isbn.nu/978-0-262-51404-0

Margulieux, L. E., Guzdial, M., & Catrambone, R. (2012). Subgoal-labeled instructional material improves performance and transfer in learning to develop mobile applications. *Proceedings of the Ninth Annual International Conference on International Computing Education Research*, 71–78. https://doi.org/10.1145/2361276.2361291

Martin, F., Lee, I., Lytle, N., Sentance, S., & Lao, N. (2020). Panel: Extending and evaluating the use-modify-create progression for engaging youth in computational thinking. *SIGCSE '20*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/SIGCSE2020Panel.pdf

Marzano, R. J., Gaddy, B. B., Foseid, M. C., Foseid, M. P., & Marzano, J. S. (2004). *Classroom management that works* [Participant handouts]. The Connecting Link.

Miao, D. Y. (2014). *PrivacyInformer: An automated privacy description generator for the MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from http://dig.csail.mit.edu/2014/Theses/ThesisDanielaMiao.pdf

Mitchell, T. M. (1997). *Machine learning*. McGraw-Hill Companies, Inc.

Morelli, R., Uche, C., Lake, P., & Baldwin, L. (2015). Analyzing year one of a CS principles project. *46th ACM Technical Symposium on Computer Science Education*, 368–373. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B3jsksMcCW5bTFBWSEZyNU5DRkk

Morelli, R., Wolber, D., Pokress, S., Turbak, F., & Martin, F. (2013). Teaching the CS principles curriculum with App Inventor (abstract only). *Proceedings of the 44th ACM Technical Symposium on Computer Science Education*, 762. https://doi.org/10.1145/2445196.2445516

Pang, H. N. (2022). *Computational action in action: Process and tools that empower students to make a real-world impact using technology* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://dspace.mit.edu/handle/1721.1/144881

Pang, H. N., Parks, R., Breazeal, C., & Abelson, H. (2022). The effect of computational action on students' software prototype ideas. *ICERI2022 Proceedings*, 1804–1814. https://doi.org/10.21125/iceri.2022.0460

Pang, H. N., Parks, R., Breazeal, C., & Abelson, H. (2023a). "how can i code A.I. responsibly?": The effect of computational action on K-12 students learning and creating socially responsible A.I. *EAAI 23 Proceedings*. Retrieved September 15, 2026, from https://ojs.aaai.org/index.php/AAAI/article/view/26902

Pang, H. N., Parks, R., Breazeal, C., & Abelson, H. (2023b). The effect of computational action on students' computational identity and self-efficacy. *EDULEARN23 Proceedings*, 8404–8412. https://doi.org/10.21125/edulearn.2023.2183

Patton, E. W. (2019a). Exploring Blockly performance: Case studies in MIT App Inventor. *Blockly User Summit 2019*. Retrieved September 15, 2026, from https://drive.google.com/open?id=1JmZ93iQOZ5LWj8kjU9AfwVEND39bwT0v

Patton, E. W. (2019b). Realtime collaboration with Blockly: Experiences with MIT App Inventor. *Blockly User Summit 2019*. Retrieved September 15, 2026, from https://drive.google.com/open?id=1Yq9_Ay4a0paFgpVQR1pJN5_wzfUtEaBW

Patton, E. W., & Deng, X. (2017). Collaborative mobile app development with MIT App Inventor [Demo presentation]. *Proceedings of Blocks & Beyond 2017*.

Patton, E. W., Druga, S., Van Brummelen, J., Mazzone, J., & Puig Fernandez, X. (2018). Panel: Block abstractions for artificial intelligence. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/blocksplus18_p14_patton.pdf

Patton, E. W., Kim, D. Y. J., Granquist, A. M., Liu, R., Scott, A., Zamanova, J., & Abelson, H. (2025). Aptly: Making mobile apps from natural language. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. https://doi.org/10.1145/3706599.3720081

Patton, E. W., Seo, A., & Turbak, F. (2019). Enhancing abstraction in App Inventor with generic event handlers. *Blocks & Beyond 2019*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/s18-patton.pdf

Patton, E. W., Sherman, M., & Tissenbaum, M. (2018). Research tools for MIT App Inventor. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/blocksplus18_p15_patton.pdf

Patton, E. W., & Tang, D. (2018). JSON interoperability in MIT App Inventor. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/blocksplus18_p13_tang.pdf

Patton, E. W., Tissenbaum, M., & Harunani, F. (2019). MIT App Inventor: Objectives, design, and development. In S. C. Kong & H. Abelson (Eds.), *Computational thinking education* (pp. 31–49). Springer Singapore. https://doi.org/10.1007/978-981-13-6528-7_3

Penrose, R., & Rindler, W. (1984). *Spinors and space-time: Vol. 1. Two-spinor calculus and relativistic fields* [Appendix: Diagrammatic notation, pp. 424–434]. Cambridge University Press. https://doi.org/10.1017/CBO9780511564048.007

Poskanzer, J. (2009). Acme license maker. Retrieved October 10, 2026, from http://www.acme.com/licensemaker/

Prutz, K. (2018). *Democratizing cross-platform mobile development with MIT App Inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from https://drive.google.com/open?id=1xCwT3fpXdAupVI5oE5_-dcmOaw0z149H

Prutz, K., & Abelson, H. (2018, May 17). *Expanding device functionality for the MIT App Inventor IoT embedded companion* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from https://drive.google.com/open?id=1AV1Za0bnoIHioj9SS0unIkU0aKT3DPBg

Rabiner, L. R. (1989). A tutorial on hidden markov models and selected applications in speech recognition. *Proceedings of the IEEE*. https://www.cs.cornell.edu/Courses/cs4758/2012sp/materials/hmm_paper_rabiner.pdf

Ravi, P., Parks, R., Masla, J., Abelson, H., & Breazeal, C. (2024). "data comes from the real world": A constructionist approach to mainstreaming K-12 data science education. *SIGCSE Virtual 2024: Proceedings of the 2024 on ACM Virtual Global Computing Education Conference*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/SIGCSE_2024_Data_Science.pdf

Ruan, L. L., Patton, E. W., & Tissenbaum, M. (2017). Evaluations of programming complexity in App Inventor. *International Conference on Computational Thinking Education 2017 (CTE*

2017). Retrieved September 15, 2026, from https://drive.google.com/open?id=0B8AUNoeCnCCARUMtS0NqN3R5d1E

Saigal, A. K. (2010). *Information accountability for mobile financial applications* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from http://dspace.mit.edu/handle/1721.1/85493

Saigal, A. K., & Saigal, A. (2011). Saathimobile and the rapid deployment of prototypes to build applications for social enterprise in the developing world. *Proceedings of the 1st International Conference on Wireless Technologies for Humanitarian Relief*, 351–356. https://doi.org/10.1145/2185216.2185311

Schiller, J., Turbak, F., Friedman, M., Abelson, H., Okerlund, J., Dominguez, J., & McKinney, A. (2014). Live programming of mobile apps in App Inventor. *Programming for Mobile and Touch (PRoMoTo)*, 1–8. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B3jsksMcCW5bS2k3TXpKVlFYX28

Schimmer, T., Hillman, G., & Stalets, M. (2018). *Standards-based learning in action: Moving from theory to practice (a guide to implementing standards-based grading, instruction, and learning)*. Solution Tree. https://isbn.nu/9781945349010

Sedgwick, R., & Wayne, K. (2011). *Algorithms* (Fourth Edition). Addison-Wesley Professional.

Seehorn, D., Carey, S., Fuschetto, B., Moix, I. L. D., O'Grady-Cunniff, D., Owens, B. B., Stephenson, C., & Verno, A. (2011). CSTA k-12 computer science standards. Retrieved August 5, 2013, from https://csta.acm.org/Curriculum/sub/K12Standards.html

Sherman, M., & Alkhatib, O. M. (2018, August 17). *MIT analysis of the 2018 Technovation student projects, sponsored by Salesforce* ("Assisted by Obada M. Alkhatib"). MIT App Inventor. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/MIT-Salesforce-Technovation-Report-2018-1-1.pdf

Shih, F., & Boortz, J. (2013). Understanding people's preferences for disclosing contextual information to smartphone apps. In L. Marinos & I. Askoxylakis (Eds.), *Human aspects of information security, privacy, and trust* (pp. 186–196). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-39345-7_20

Shih, F., & Li, W. (2013). Unlocking smartphone data for educational use in teaching and learning environment. *Smart University Workshop, Eighth International and Interdisciplinary Conference on Modeling and Using Context*. Retrieved September 15, 2026, from https://appinventor.mit.edu/explore/sites/explore.appinventor.mit.edu/files/resources/smartUniv.pdf

Shih, F., Seneviratne, O., Miao, D., Liccardi, I., Kagal, L., Patton, E. W., Meier, P., & Castillo, C. (2013). Democratizing mobile app development for disaster management. *AIIP Workshop on Artificial Intelligence Problems and Approaches for Intelligent Environments and Workshop on Semantic Cities*, 39–42. Retrieved September 15, 2026, from http://dig.csail.mit.edu/2013/QCRI-DIG-project/paper.pdf

Shone, J. (2022). *Yes, you can make an app too: A systematic study of prompt engineering in the automatic generation of mobile applications from user queries* (Research Paper). Research Science Institute. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Shone_Jasmine_Rachel_RSIFinal.pdf

Shone, J., Liu, R., Patton, E., & Kim, D. Y. J. (2022). Exploring prompt engineering for generative AI-based app generation. *Acta Scientific Computer Sciences*, *5*(5). Retrieved September 15, 2026, from https://actascientific.com/ASCS/ASCS-05-0448.php

Smith, R. (2004). *Conscious classroom management: Unlocking the secrets of great teaching*. Conscious Teaching Publications.

Snyder, L., Astrachan, O., & Briggs, A. (2013). Seven big ideas of computer science. Retrieved August 5, 2013, from https://csprinciples.cs.washington.edu/sevenbigideas.html

Stephenson, C. (2012, November 15). The national imperative for k–12 computer science education. Retrieved August 20, 2013, from https://www.csta.acm.org/Advocacy_Outreach/sub/PresentationFiles/MicrosoftPanel_Stephenson.pdf

Tang, D. (2019). *Empowering novices to understand and use machine learning with personalized image classification models, intuitive analysis tools, and MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=1lfOVkajhYZaFtVGZnNWjQn6nBrBLWzE1

Tang, D., Utsumi, Y., & Lao, N. (2019). PIC: A personal image classification webtool for high school students. *IJCAI 2019 EduAI Workshop*. Retrieved September 15, 2026, from https://drive.google.com/open?id=18x1pGEoKrq_4ShkFdH2hYjoTtc0qyECa

Tang, L. (2017). *Creating customized user interfaces for App Inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B8AUNoeCnCCAN2ZaeFRHcENWbGM

Tang, L. (2018). *Constrained sets: The effects of multi-layered environments in learning App Inventor* [Master's thesis, Massachusetts Institute of Technology] [Advisor: Harold Abelson]. Retrieved September 15, 2026, from https://dspace.mit.edu/handle/1721.1/119707

Taylor, E. F., Vokos, S., O'Meara, J. M., & Thornber, N. S. (1998). Teaching Feynman's sum-over-paths quantum theory. *Computer in Physics*, *12*(2), 190–199. https://doi.org/10.1063/1.168652

This is the title of the webpage. (2009, February). Retrieved August 29, 2013, from http://www.example.com

Tian, M. C. (2023). *The design and implementation of stateful ConvoBlocks* [BA thesis]. Wellesley College. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/WCTC_2023_TianMingyan_TheDesignAndImp.pdf

Tissenbaum, M., Sheldon, J., & Abelson, H. (2018). Reducing the barriers for computational action: How blocks-based programming can support digital empowerment. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/blocksplus18-paper18.pdf

Tissenbaum, M., Sheldon, J., & Abelson, H. (2019). From computational thinking to computational action. *Communications of the ACM*, *62*(3), 34–36. https://doi.org/10.1145/3265747

Tissenbaum, M., Sheldon, J., Patton, E. W., Gupta, A., Zhang, E., & Gopinath, D. (2017). Off the screen, and into the world of everyday objects: Computational thinking for youth with the internet of things. *International Conference on Computational Thinking Education 2017 (CTE 2017)*. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B8AUNoeCnCCAOUx2NFU5UzExY1E

Tissenbaum, M., Sheldon, J., Seop, L., Lee, C., & Lao, N. (2017). Critical computational empowerment: Engaging youth as shapers of the digital future. *Global Engineering Education Conference (EDUCON), 2017 IEEE*, 1705–1708. https://doi.org/10.1109/EDUCON.2017.7943078

Toffoli, T., & Margolus, N. (1987). *Cellular automata machines: A new environment for modeling*. The MIT Press.

Tsai, S. (2016). *Understanding App Inventor forums* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B3jsksMcCW5bS1Z4VkFmaGRCY1E

Tseng, C. H., Tissenbaum, M., Kuan, W. H., Hsu, F. C., & Wong, C. C. (2018). A design-based approach to implementing a computational thinking curriculum with App Inventor and the internet of things. *International Conference on Computational Thinking Education 2018*

(CTE 2018). Retrieved September 15, 2026, from https://drive.google.com/open?id=1P-jl1gNdkFIIxEeMdo9qiNcvdaLJbxnl

Tucker, A. (2002). *Applied combinatorics* (Fourth). John Wiley & Sons. https://isbn.nu/047143809X

Turbak, F., Sandu, S., Kotsopoulos, O., Erdman, E., Davis, E., & Chadha, K. (2012). Blocks languages for creating tangible artifacts. *IEEE Symposium on Visual Languages and Human-Centric Computing*, 137–144. Retrieved September 15, 2026, from http://cs.wellesley.edu/~fturbak/pubs/VLHCC-2012-paper-turbak.pdf

Turbak, F., Sherman, M., Martin, F., Wolber, D., & Crawford Pokress, S. (2014). Events-first programming in App Inventor. *Journal of Computing Sciences in Colleges*, *29*(6), 81–89. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B3jsksMcCW5bSGhUUjctMkFXT1E

Van Brummelen, J., & Abelson, H. (2018). "what's conversational AI?" with MIT App Inventor and Amazon Alexa [Poster]. *Amazon Research Days 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/explore/sites/explore.appinventor.mit.edu/files/Amazon_Alexa_Day_Poster_portrait.pdf

Van Brummelen, J. (2019a). Conversational agents to democratize artificial intelligence. *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 239–240. https://doi.org/10.1109/VLHCC.2019.8818805

Van Brummelen, J. (2019b). Conversational artificial intelligence development tools for K-12 education. *AAAI Fall Symposium*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/JessVBPublications/AAAI_Fall_Symposium_2019_SMConvAI_Final.pdf

Van Brummelen, J. (2019c). *Tools to create and democratize conversational artificial intelligence* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=1AMNP3hvfS1EEgnwFugJkIjEpQ3KMtFQk

Van Brummelen, J. (2022). *Empowering K-12 students to understand and design conversational agents: Concepts, recommendations and development platforms* [Doctoral dissertation, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://dspace.mit.edu/entities/publication/b678c11f-1069-48b3-98e5-fe2de2be2681

Van Brummelen, J., Heng, T., & Tabunshchyk, V. (2021). Teaching tech to talk: K-12 conversational artificial intelligence literacy curriculum and development tools. *Proceedings of the AAAI Conference on Artificial Intelligence*, *35*(17), 15655–15663. https://doi.org/10.1609/aaai.v35i17.17844

Van Brummelen, J., Kelleher, M., Tian, M. C., & Nguyen, N. H. (2023). What do children and parents want and perceive in conversational agents? towards transparent, trustworthy, democratized agents. *Proceedings of the 22nd Annual ACM Interaction Design and Children Conference*, 187–197. https://doi.org/10.1145/3585088.3589353

Van Brummelen, J., Shen, J. H., & Patton, E. W. (2019). The popstar, the poet, and the grinch: Relating artificial intelligence to the computational thinking framework with block-based coding. *International Conference on Computational Thinking Education 2019 (CTE2019)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/JessVBPublications/Popstar_Poet_Grinch_CTE2019.pdf

Van Brummelen, J., Tabunshchyk, V., & Heng, T. (2021). "Alexa, can i program you?": Student perceptions of conversational artificial intelligence before and after programming Alexa. *Proceedings of the 20th Annual ACM Interaction Design and Children Conference*, 305–313. https://doi.org/10.1145/3459990.3460730

Van Brummelen, J., Tian, M. C., Kelleher, M., & Nguyen, N. H. (2023). Learning affects trust: Design recommendations and concepts for teaching children — and nearly anyone — about

conversational agents. *Proceedings of the AAAI Conference on Artificial Intelligence*, *37*, 15860–15868. Retrieved September 15, 2026, from https://arxiv.org/pdf/2209.05063.pdf

Van Brummelen, J., Weng, K., Lin, P., & Yeo, C. (2020). CONVO: What does conversational programming need? *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 1–5. https://doi.org/10.1109/VL/HCC50065.2020.9127277

Van Brummelen, J., Yeo, C., & Weng, K. (2020). Learning to program conversationally: A conversational agent to further democratize programming. *14th International Technology, Education and Development Conference (INTED2020)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/JessVBPublications/INTED2020_abstract.pdf

van de Donk, W., Loader, B. D., Nixon, P. G., & Rucht, D. (Eds.). (2004). *Cyberprotest: New media, citizens and social movements*. Routledge. Retrieved October 29, 2013, from www.jstor.com/somearticle.html

Wang, K. (2018). *Creating a database log history page for cloud components in App Inventor to support software development education* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from https://drive.google.com/open?id=0BzGL9U0OLqlkQUdxQ281Z3B6UlA4VE42UzVNZ1hGb0pKUkx3

Watts, A. (1972). *The supreme identity*. Vintage Books.

Weng, K. (2020). *Examining conversational programming design needs with convo, a voice-first conversational programming system using natural language* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/MEng_Thesis_Kevin_Weng.pdf

Wikibooks contributors. (2018, November 27). Java programming / primitive types — Wikibooks, the free textbook project. Retrieved February 18, 2019, from https://en.wikibooks.org/w/index.php?title=Java_Programming/Primitive_Types&oldid=3491137

Wikipedia contributors. (2004, July 22). Plagiarism — Wikipedia, the free encyclopedia. Retrieved July 27, 2004, from https://en.wikipedia.org/w/index.php?title=Plagiarism&oldid=5139350

Wikipedia contributors. (2010a, June 7). Permutation — Wikipedia, the free encyclopedia. Retrieved October 26, 2010, from https://en.wikipedia.org/w/index.php?title=Permutation&oldid=366563313

Wikipedia contributors. (2010b, June 14). Combination — Wikipedia, the free encyclopedia. Retrieved October 26, 2010, from https://en.wikipedia.org/w/index.php?title=Combination&oldid=367895591

Wikipedia contributors. (2010c, June 19). Combinatorics — Wikipedia, the free encyclopedia. Retrieved October 26, 2010, from https://en.wikipedia.org/w/index.php?title=Combinatorics&oldid=368950754

Wikipedia contributors. (2013, May 2). Hungarian algorithm — Wikipedia, the free encyclopedia. Retrieved June 17, 2013, from https://en.wikipedia.org/w/index.php?title=Hungarian_algorithm&oldid=553176136

Wikipedia contributors. (2017a, July 29). Signedness — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Signedness&oldid=792830866

Wikipedia contributors. (2017b, August 13). Infix notation — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Infix_notation&oldid=795311885

Wikipedia contributors. (2018a, December 6). Russell's paradox — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Russell's_paradox&oldid=872268102

Wikipedia contributors. (2018b, December 8). Ones' complement — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Ones'_complement&oldid=872645202

Wikipedia contributors. (2018c, December 16). NAND logic — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=NAND_logic&oldid=874024831

Wikipedia contributors. (2018d, December 19). Fixed-point arithmetic — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Fixed-point_arithmetic&oldid=874423261

Wikipedia contributors. (2018e, December 23). Multiplication algorithm — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Multiplication_algorithm&oldid=875028575

Wikipedia contributors. (2019a, January 5). Unary operation — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Unary_operation&oldid=876888704

Wikipedia contributors. (2019b, January 7). Complement (set theory) — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Complement_(set_theory)&oldid=877236947

Wikipedia contributors. (2019c, January 10). Positional notation — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Positional_notation&oldid=877761722

Wikipedia contributors. (2019d, January 14). Naive set theory — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Naive_set_theory&oldid=878461198

Wikipedia contributors. (2019e, January 23). Adder (electronics) — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Adder_(electronics)&oldid=879794929

Wikipedia contributors. (2019f, January 28). Integer overflow — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Integer_overflow&oldid=880564879

Wikipedia contributors. (2019g, January 28). Polish notation — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Polish_notation&oldid=880582128

Wikipedia contributors. (2019h, January 30). Bit — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Bit&oldid=880961230

Wikipedia contributors. (2019i, February 3). Functional completeness — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Functional_completeness&oldid=881651954

Wikipedia contributors. (2019j, February 5). Venn diagram — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Venn_diagram&oldid=881835160

Wikipedia contributors. (2019k, February 7). NOR logic — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=NOR_logic&oldid=882243128

Wikipedia contributors. (2019l, February 8). Bitwise operation — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Bitwise_operation&oldid=882396419

Wikipedia contributors. (2019m, February 9). Two's complement — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Two's_complement&oldid=882476598

Wikipedia contributors. (2019n, February 12). Binary operation — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Binary_operation&oldid=882927455

Wikipedia contributors. (2019o, February 12). IEEE 754 — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=IEEE_754&oldid=883044346

Wikipedia contributors. (2019p, February 13). Nibble — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Nibble&oldid=883088049

Wikipedia contributors. (2019q, February 15). Division algorithm — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Division_algorithm&oldid=883467096

Wikipedia contributors. (2019r, February 17). Barber paradox — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Barber_paradox&oldid=883712357

Wikipedia contributors. (2019s, February 18). Boolean algebra — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Boolean_algebra&oldid=883953881

Wikipedia contributors. (2019t, February 18). Floating-point arithmetic — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Floating-point_arithmetic&oldid=883934950

Wikipedia contributors. (2019u, February 18). Logic gate — Wikipedia, the free encyclopedia. Retrieved February 18, 2019, from https://en.wikipedia.org/w/index.php?title=Logic_gate&oldid=883992939

Wikipedia contributors. (2023, April 26). Carmichael number — Wikipedia, the free encyclopedia. Retrieved August 13, 2023, from https://en.wikipedia.org/w/index.php?title=Carmichael_number&oldid=1151790558

Wikipedia contributors. (2026a, September 14). Quantum nonlocality — Wikipedia, the free encyclopedia [Relevant section: #Einstein,_Podolsky_and_Rosen]. Retrieved October 2, 2026, from https://en.wikipedia.org/w/index.php?title=Quantum_nonlocality&oldid=1374827206

Wikipedia contributors. (2026b, October 10). Wikipedia — Wikipedia, the free encyclopedia. Retrieved October 11, 2026, from https://en.wikipedia.org/w/index.php?title=Wikipedia&oldid=1379559220

Wing, J. M. (2006). Computational thinking. *Communications of the ACM*, *49*(3). https://www.cs.cmu.edu/~wing/publications/Wing06.pdf

Wolber, D., Abelson, H., & Friedman, M. (2014). Democratizing computing with App Inventor. *GetMobile: Mobile Computing and Communications*, *18*(4), 53–58. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B3jsksMcCW5baTlwSEpVWjBlQ00

Xi, W., & Patton, E. W. (2018). A blocks-based approach to internet of things in MIT App Inventor. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/blocksplus18_p12_xi.pdf

Xie, B. (2016). *Progression of computational thinking skills demonstrated by App Inventor users* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=0B3jsksMcCW5bZGVuemF5VDQ0QWc

Xie, B., Shabir, I., & Abelson, H. (2015). Measuring the usability and capability of App Inventor to create mobile applications. *PROMOTO 2015*, 108. Retrieved September 15, 2026, from http://dspace.mit.edu/handle/1721.1/98913

Yoo, D., Kim, D. Y. J., & Lopes, E. (2023). Digital art therapy with gen AI: Mind palette. *11th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/yoo2023digital.pdf

Yu, J. (2021). *Using facemesh in MIT App Inventor to empower students to apply artificial intelligence* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=1-QBYzYf5uReYSCT6f5OOI2VPDijXggE-

Yuan, J. (2025, May). *Empowering mobile-only app generation — Offline AI code generation with App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Joyce2025Thesis.pdf Thesis supervisor: Harold Abelson, Class of 1922 Professor of Computer Science and Engineering. Accepted by: Katrina LaCurts, Chair, Master of Engineering Thesis Committee.

Zamanova, J., Patton, E. W., & Abelson, H. (2026). *Visual project comparison system for block-based graphical programming platforms* [Submitted].

Zhang, F., Shih, F., & Weitzner, D. (2013). No surprises: Measuring intrusiveness of smartphone applications by detecting objective context deviations. *Proceedings of the 12th ACM Workshop on Workshop on Privacy in the Electronic Society*, 291–296. https://doi.org/10.1145/2517840.2517864

Zhou, A., Kim, D. Y. J., & Abelson, H. (2023). Learning and using image classifiers by creating mobile applications. *INTED2023 Proceedings*. Retrieved September 15, 2026, from https://docs.google.com/document/d/1f0LKoJKpiF--_DlnXU0pIEnANthFSjyj-Z1MNkte3m4/edit

Zhou, X., Van Brummelen, J., & Lin, P. (2020). *Designing AI learning experiences for K-12: Emerging works, future opportunities and a design framework* [Unpublished preprint; no subsequent peer-reviewed venue found as of 2026]. Retrieved September 15, 2026, from https://arxiv.org/pdf/2009.10228.pdf

Zhu, J. (2021). *Creating your own conversational artificial intelligence agents using convo, a conversational programming system* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/JessicaZhu_MEng_Thesis.pdf

Zhu, J., & Van Brummelen, J. (2021). Teaching students about conversational AI using convo, a conversational programming agent. *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 1–5. https://doi.org/10.1109/VL/HCC51201.2021.9576290

Zhu, K. (2019). *An educational approach to machine learning with mobile applications* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=1Q0D2wyiPLfpZzFc0cxNk5jl9PWuN3oYl

Zurek, W. H. (2002). Decoherence and the transition from quantum to classical—Revisited. *Los Alamos Science*, *27*. Retrieved October 7, 2026, from https://arxiv.org/abs/quant-ph/0306072